AS YOU SIT DOWN

# Get your workspace open

THREE THINGS

- VS Code open on your **ECON209** folder
- Your Problem Set 1 notebook
- Your **README.md**

IF SOMETHING ISN'T WORKING

- Sit next to someone whose setup runs
- Follow along now, fix yours after

APPLIED ECONOMETRICS · RICE UNIVERSITY

# Lab Four

Submitting your homework, organizing and annotating code, and an introduction to Git and GitHub.


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Questions, any time

**Sewall 207**
Mondays

TONIGHT

# Four parts

1. **Problem Set 1:** what tripped people up when submitting
2. **Organizing and annotating** a homework notebook, with an example
3. **What the HTML holds**, and what to submit
4. **Git and GitHub:** what they are, and how to work on a project with a partner

PART ONE · PROBLEM SET 1

# What tripped people up

### EXPORTING

- Couldn't find Export
- Or Export gave an error

### WHAT TO INCLUDE

- The data file?
- The codebook?
- The zip, or the whole folder?

### WHICH FILE

- The `.ipynb` uploaded instead of the `.html`

We'll fix each of these tonight. First, what the notebook itself should look like.

# How a homework notebook should read

### MARKDOWN CELLS

- **At the top:** a title, your name, and what the notebook does
- **Before each code cell:** what the next cell does
- **After each result:** your answer, in your words — not as a comment in the code

### CODE CELLS

- At least one cell for each part of a question: 1a, 1b, ...
- One or two outputs per cell, at most
- A comment on every line, for someone who has never coded

Someone should be able to follow it from top to bottom without running anything.

# One step: a markdown cell, then the code

## MARKDOWN CELL

Step 5 · Descriptive statistics by state

The next cell splits the stores into New Jersey and Pennsylvania and, for each of our three variables, reports count, mean, std, min and max.

## CODE CELL, THE NEXT ONE DOWN

```
# The three variables we want to describe.
columns = ["FTE", "WAGE_ST", "PMEAL"]

# Read this chain of steps from top to bottom:
#   .groupby("state")  split the stores into NJ and PA
#   [columns]          keep only our three variables
#   .agg([...])        compute these five statistics for each
#   .round(2)          round to two decimal places
#   .T                 flip the table: variables down the side,
#                      the two states across the top
summary = (
    df.groupby("state")[columns]
    .agg(["count", "mean", "std", "min", "max"])
    .round(2)
    .T
)

# Show the table.
summary
```

# The table, then what it means

| state | | NJ | PA |
|---|---|---|---|
| FTE | count | 321.00 | 77.00 |
| | mean | 20.44 | 23.33 |
| | std | 9.11 | 11.86 |
| | min | 5.00 | 7.50 |
| | max | 85.00 | 70.50 |
| WAGE_ST | count | 314.00 | 76.00 |
| | mean | 4.61 | 4.63 |
| | std | 0.35 | 0.35 |
| | min | 4.25 | 4.25 |
| | max | 5.75 | 5.50 |
| PMEAL | count | 311.00 | 76.00 |
| | mean | 3.35 | 3.04 |
| | std | 0.64 | 0.60 |
| | min | 2.50 | 2.28 |
| | max | 5.86 | 5.40 |

WHAT WE FOUND · MARKDOWN, EXCERPT

- The average New Jersey store had 20.4 full-time-equivalent employees, compared with 23.3 in Pennsylvania — about 2.9 fewer.
- Starting wages were almost the same in the two states: 4.61 dollars an hour in New Jersey and 4.63 in Pennsylvania.
- A full meal cost about 31 cents more in New Jersey: 3.35 dollars compared with 3.04.

On a problem set, this last cell is the part you write yourself.

# What the HTML holds

YOUR HOMEWORK FOLDER

```
homework/
├── data/
│   └── njmin/
│       └── public.dat
├── code/
│   └── ps1.ipynb
└── output/
    └── ps1.html
```

IN THE HTML

- Your code and your markdown notes
- The outputs from the last time you ran it

NOT IN THE HTML

- `public.dat`, the codebook, the zip

Like saving a Word document as a PDF: a copy of what is on the page.

PART THREE · WRITTEN ANSWERS

# Where your written answers go

### OPTION 1 · IN THE NOTEBOOK

- A markdown cell for each written question, headed with its number: `## I.1a`
- It exports with everything else, in the same HTML
- Formulas go between $ signs: `$P(z_1 = 1 \mid NJ = 1)$`

### OPTION 2 · A SEPARATE PDF

- Fine to upload alongside the HTML
- Clearly organized: every answer labeled with its question number, in order

Either way, written answers are yours: no AI for non-coding questions.

# Export and submit

1. Run All — the double arrow at the top of the notebook
2. Save — Cmd/Ctrl + S
3. "…" → Export → HTML — the menu at the top of the notebook
4. Save it where you'll find it — `output/` is a good home
5. Open the .html in your browser. Is every table there?
6. Upload the .html to Canvas, and your written PDF if you have one

The .ipynb needs Jupyter to open. The .html opens in any browser, which is why it's the file you submit.

PART FOUR · GIT

# Git: version control on your laptop

- Tracks changes to the files in a folder
- A **commit** is a snapshot you choose to take, with a short message
- Each commit records exactly what changed, when, and who
- You can go back to any earlier snapshot
- All on your laptop; it works offline

AN EXAMPLE HISTORY

```
Sep 14  Load public.dat
Sep 15  Add FTE and meal price
Sep 16  Add summary table
Sep 17  Write interpretation
```

One commit per line. Each is a snapshot you can return to.

PART FOUR · GITHUB

# GitHub: your history, online

YOUR FOLDER

The files you edit

GIT

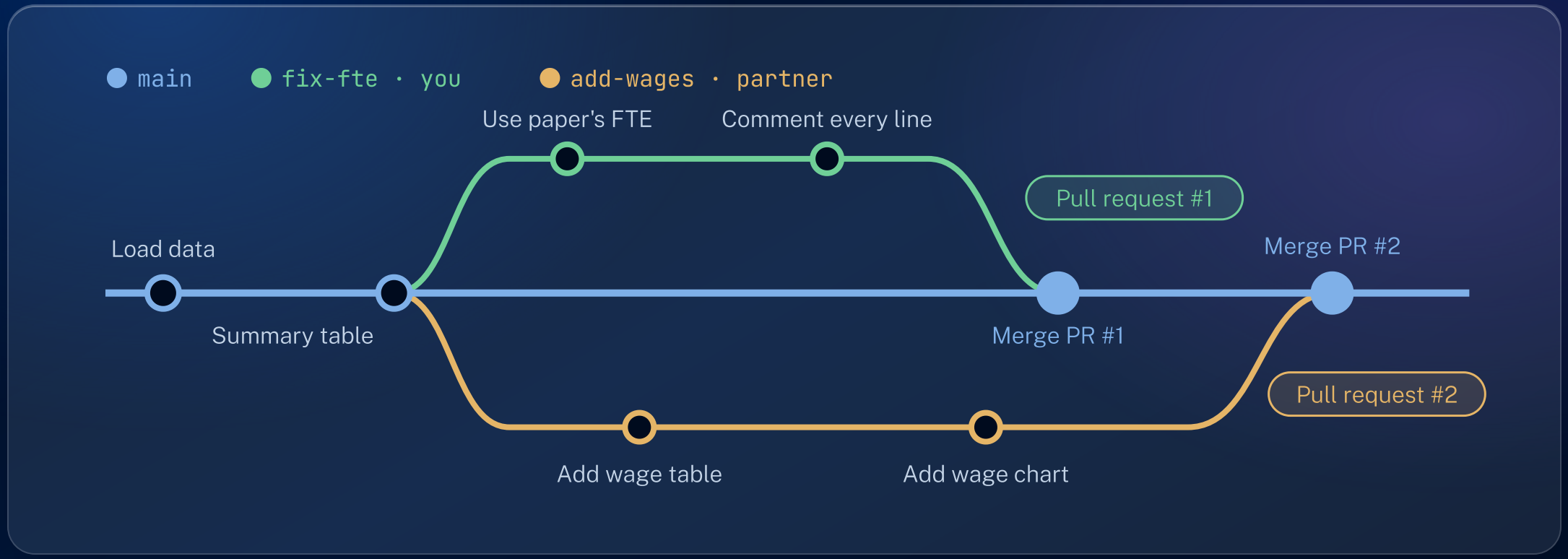
Its history, on your laptop

GITHUB

A copy of that history, online

- A project on GitHub is called a **repository**
- **Push** sends your commits up; **pull** brings down commits you don't have
- Open it from any computer; a repository can be private or public

# Two people, one project




Each dot is a commit. You each work on your own branch at the same time, and each branch comes back into `main` through a pull request.

PART FOUR · PULL REQUESTS

# What a pull request is

A PROPOSAL TO MERGE CHANGES INTO A PROJECT

- Teammates see the exact lines that changed, and comment on them
- Nothing changes in `main` until it's merged
- **The why is only there if you write it:** say why in your message

WHAT A REVIEWER SEES

Use the paper's FTE definition
Jessica · Sep 16

```
- df["FTE"] = df["EMPFT"] + df["EMPPT"]
+ df["FTE"] = (df["EMPFT"] + df["NMGRS"]
+               + 0.5 * df["EMPPT"])
```

One line removed (red), two added (green), and a message saying why.

PART FOUR · TWO WORDS THAT SOUND ALIKE

# Pull vs. pull request

PULL

`GitHub → your laptop`

- Brings down commits you don't have yet
- Your copy catches up with everyone else's
- In VS Code: **Sync Changes**

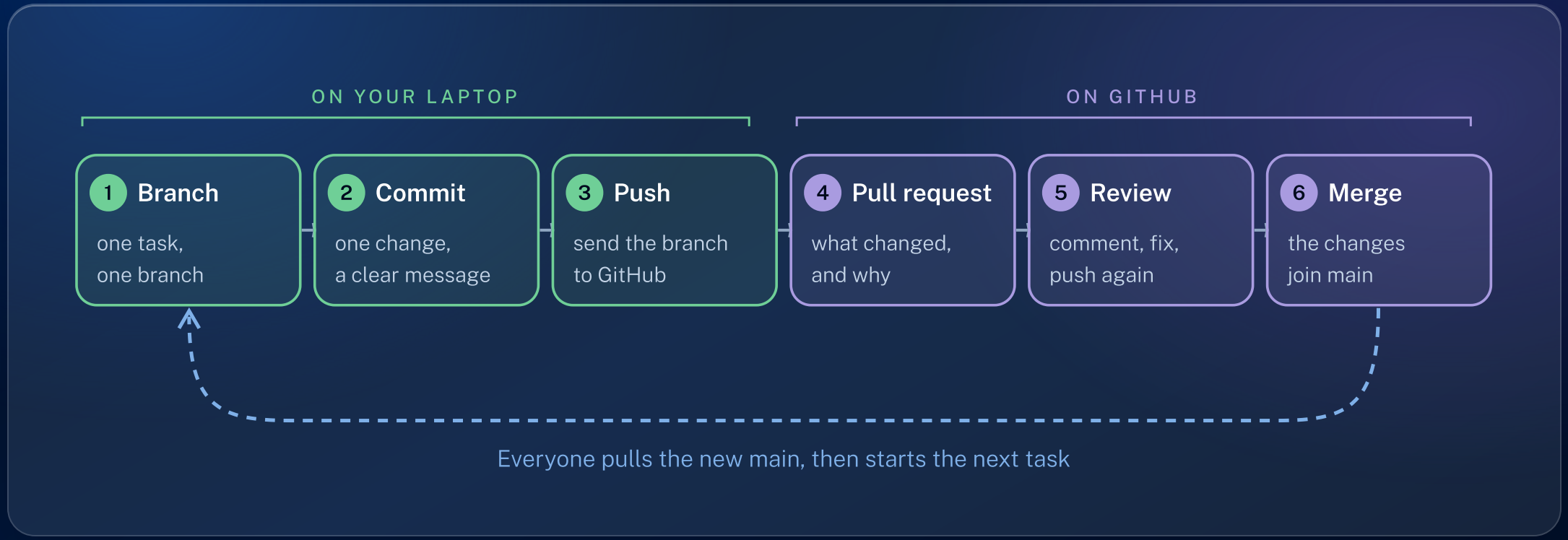
PULL REQUEST

`your branch → main`

- Asks for your branch to be merged into `main`
- Happens on GitHub, with a review first
- You're asking the project to *pull in* your changes

PART FOUR · GITHUB FLOW

# One task, start to finish




If you both changed the same line, that's a **merge conflict**: Git can't choose, so you decide which version to keep before it can merge.

STEP BY STEP · YOUR SIDE

# Proposing a change

1. Start from the latest main. On `main`, click Sync Changes
2. Make a branch. Branch name, bottom left → Create new branch
3. Commit as you work. A message that says what and why → Commit
4. Send the branch to GitHub. Click Publish Branch
5. Open the pull request. On GitHub: Compare & pull request → base `main`, compare your branch → say what and why → Create pull request

STEP BY STEP · YOUR FRIEND'S SIDE

# Reviewing a pull request

1. **Find it.** On the repository page, click **Pull requests**, then click theirs
2. **Read the changes.** Click **Files changed**: removed lines in red, added lines in green
3. **Comment on a line.** Hover over it and click the blue comment icon
4. **Decide.** Click **Review changes**, choose one, then **Submit review**

THE THREE CHOICES

- **Approve:** ready to merge
- **Request changes:** they fix it and push again; the pull request updates
- **Comment:** feedback, no decision yet

STEP BY STEP · FINISHING

# Merging, then everyone pulls

1. **Merge.** At the bottom of the pull request, click **Merge pull request**, then **Confirm merge**
2. **Tidy up.** Delete the branch, if GitHub offers
3. **Everyone pulls.** Each of you switches to `main` (branch name, bottom left), then clicks **Sync Changes**

WHAT CAN STOP A MERGE

- **A merge conflict:** you both changed the same lines. Decide which version to keep first
- **A draft pull request:** mark it ready for review first

BEFORE YOU GO

# Three things

- [ ] Add tonight's rules to your `README.md`: a markdown cell before each code cell, a code cell for each part of a question, a comment on every line, and answers in markdown, in your own words.
- [ ] **Coming up:** putting your project folder on GitHub.
- [ ] **Problem Set 2** posts Oct 8.

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